RNA bases and base pairs as circuits

Infrared absorption and release of the four RNA bases, the Watson–Crick pairs and the G·U wobble pair from kronspice's AC analysis — doc/chemistry-plan.md

kronspice simulation report · 2026-11-07 · AMBER RNA OL3, harmonic, gas phase

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Summary

The RNA twin of dna_vibration. The four RNA bases (methyl-capped), the Watson–Crick pairs A·U and G·C and RNA's wobble pair G·U are built as electric circuits with the mobility analogy: atoms are capacitors (mass), every bond and angle of the AMBER force field is a spring inductor, and Kron's connection tensor joins the bond frame to the atom frame. Light enters as a current source at every atom. kronspice's AC analysis gives the infrared light each molecule absorbs, and two independent routes give the light it releases. Stacks of 2–4 pairs are placed on the A-form step of the NMR duplex 1EKA.

- **Starting structure:** PDB 1EKA, the solution NMR structure of r(GAGUGCUC)$_2$ with all hydrogens (Chen et al., Biochemistry 39, 8970, 2000): A·U = A2·U7, G·C = G1·C8, G·U = G5·U4; everything minimized in Go.
- **All seven molecules are true minima** (RMS gradient ≤ 7.0×10^{-10} kcal/mol/Å, all 3N−6 vibrations real) and electrically small (d/λ = 2.0×10^{-4} to 4.3×10^{-4}).
- **Thermal infrared release at 300 K** is 1.24×10^{-20} W (U) to 5.16×10^{-20} W (G·C) per molecule; the wobble pair G·U gives 4.13×10^{-20} W.
- **N–H stretches sit 3.4–6.0 % below the measurements** (harmonic AMBER against anharmonic data); uracil's C2=O comes out +3.0 %, its C4=O has no clean line (best mode 1458.0 cm^{-1}, 19 % of the bond's energy, −14.2 %).
- **The wobble pair keeps its two hydrogen bonds** (G N1–H1···O2 U, U N3–H3···O6 G) and its stiffest pairing mode, 131 cm^{-1}, sits beside A·U's 134 cm^{-1} and below G·C's 152 cm^{-1}.
- **Pairing red shifts are not reproduced:** with fixed charges the H-bonded stretches move up on pairing.

The model

- **Molecules.** 9-methyladenine, 1-methyluracil, 9-methylguanine and 1-methylcytosine (the methyl stands in for the ribose) and the pairs A·U, G·C and G·U, cut from PDB 1EKA and minimized in the AMBER RNA force field OL3 (Cornell et al. base terms, bsc0, OL3 χ; OpenMM's amber14/RNA.OL3.xml). prep/prep.py writes molecules.json; the methyl caps are made exactly as in dna_vibration.
- **Circuit** (circuit.go, as in dna_vibration). Each AMBER bond and angle is a spring branch (VCVS chain for Wilson's B row, an inductor for the spring force, CCCS back to the atoms); everything else is the residual stiffness R = K − Σ k·B^{T}B through displacement integrators and one VCCS per non-zero entry. Forces between the two bases of a pair form the group G_{HB}; in a stack, forces between bases of different pairs form G_{ST}.
- **Outputs.** Absorption is the power the field puts in during the AC sweep (0–3700 cm^{-1}, 2 cm^{-1} steps, averaged over x, y, z). Release per mode comes from the Einstein A coefficient and from Maxwell's radiated power of the vibrating dipole (em.LarmorPower); they must agree. Release spectra are given for one quantum per mode and for thermal equilibrium at 300 K.
- **Stacks.** 1EKA has no AA/UU, GG/CC or CG step, so stacks repeat its real GA/UC step (G1·C8 → A2·U7) in a simplified base-pair frame (Olson et al. 2001); U·A and C·G pairs are A·U and G·C turned 180° about the dyad. The tandem wobble stack GU2 is cut from 1EKA as it is.

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Results

Bases and pairs (run_output.txt)

molecule	atoms	vibrations	range [cm^{-1}]	RMS gradient [kcal/mol/Å]	springs	pairing sources	d/λ	thermal release at 300 K
A	18	48	48–3367	3.8×10^{-13}	49	0	2.5×10^{-4}	2.52×10^{-20} W
U	15	39	52–3313	1.6×10^{-13}	39	0	2.0×10^{-4}	1.24×10^{-20} W
G	19	51	31–3363	5.0×10^{-12}	52	0	2.5×10^{-4}	2.97×10^{-20} W
C	16	42	45–3366	2.1×10^{-13}	41	0	2.2×10^{-4}	2.52×10^{-20} W
A·U pair	33	93	12–3365	7.5×10^{-13}	88	5157	4.3×10^{-4}	3.60×10^{-20} W
G·C pair	35	99	18–3370	4.7×10^{-11}	93	5787	4.2×10^{-4}	5.16×10^{-20} W
G·U pair	34	96	12–3363	7.0×10^{-10}	91	5436	4.3×10^{-4}	4.13×10^{-20} W

A = 9-methyladenine, U = 1-methyluracil, G = 9-methylguanine, C = 1-methylcytosine. Against dna_vibration: the A, G and C N–H lines agree with the DNA run to within half a wavenumber (A NH_2 antisymmetric 3367.0 against 3367.1 cm^{-1}); uracil releases 1.24×10^{-20} W against thymine's 1.28×10^{-20} W, and A·U 3.60×10^{-20} W against A·T's 3.68×10^{-20} W.

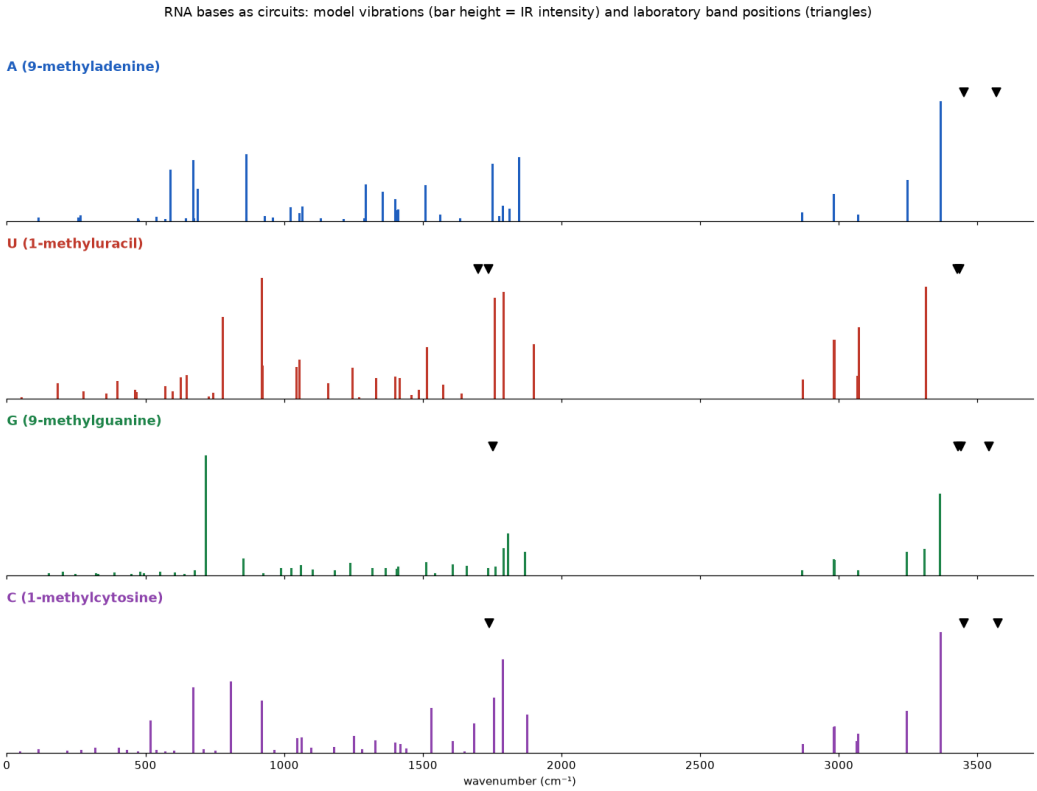


Figure 1. Model lines of the four RNA bases (bar height: IR intensity) with the laboratory band positions (triangles).

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Model against measurement (summary/measured.csv)

band	measured [cm^{-1}]	model [cm^{-1}]	difference	conditions
A NH_2 antisymmetric stretch	3569	3367.0	−5.7 %	9H-adenine, gas, jet-cooled IR–UV
A NH_2 symmetric stretch	3452	3246.7	−5.9 %	9H-adenine, gas, jet-cooled IR–UV
U N3–H stretch	3428.6	3312.7	−3.4 %	1-methyluracil, Ar matrix 12 K, main band (site component 3433.4)
U N3–H stretch (uracil, gas)	3436	3312.7	−3.6 %	uracil (not methylated), hot gas 325 °C, FT absorption/emission
U C2=O stretch	1737.5	1789.3	+3.0 %	1-methyluracil, Ar matrix 12 K (quoted in the text)
U C4=O stretch	1700	1458.0	−14.2 %	1-methyluracil, Ar matrix 12 K (quoted in the text)
G NH_2 antisymmetric stretch	3543	3363.3	−5.1 %	9-methylguanine amino-oxo, Ar matrix 14 K (site 3543/3535)
G N1–H stretch	3440	3308.3	−3.8 %	9-methylguanine amino-oxo, Ar matrix (3440/3435)
G NH_2 symmetric stretch	3430	3244.0	−5.4 %	9-methylguanine amino-oxo, Ar matrix (3430/3427)
G C6=O stretch	1752	1544.2	−11.9 %	9-methylguanine amino-oxo, Ar matrix (1752/1741)
C NH_2 antisymmetric stretch	3572.7	3365.6	−5.8 %	cytosine keto form, helium nanodroplets 0.37 K
C NH_2 symmetric stretch	3451.7	3244.6	−6.0 %	cytosine keto form, helium nanodroplets
C C2=O stretch	1740	1685.2	−3.1 %	cytosine keto, gas GC-FTIR 300 °C (as quoted)

Adenine, guanine and cytosine values and sources are reused from dna_vibration (the same molecules); the uracil values are new: 1-methyluracil in an argon matrix (Ivanov et al. 2002) and uracil in the gas phase (Colarusso et al. 1997). No gas-phase or droplet IR spectrum of an A·U or G·U pair, and no measured low-frequency mode of an A-form RNA duplex, was found, so neither is compared.

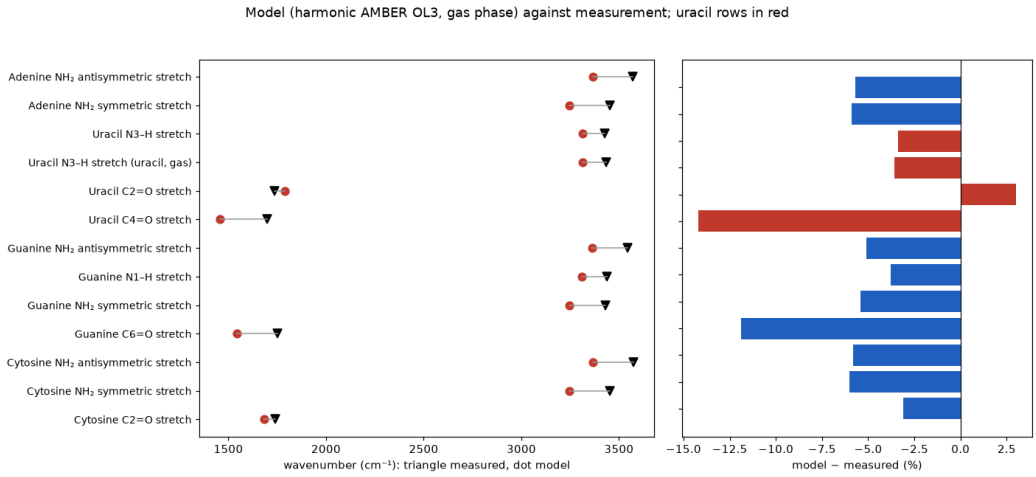


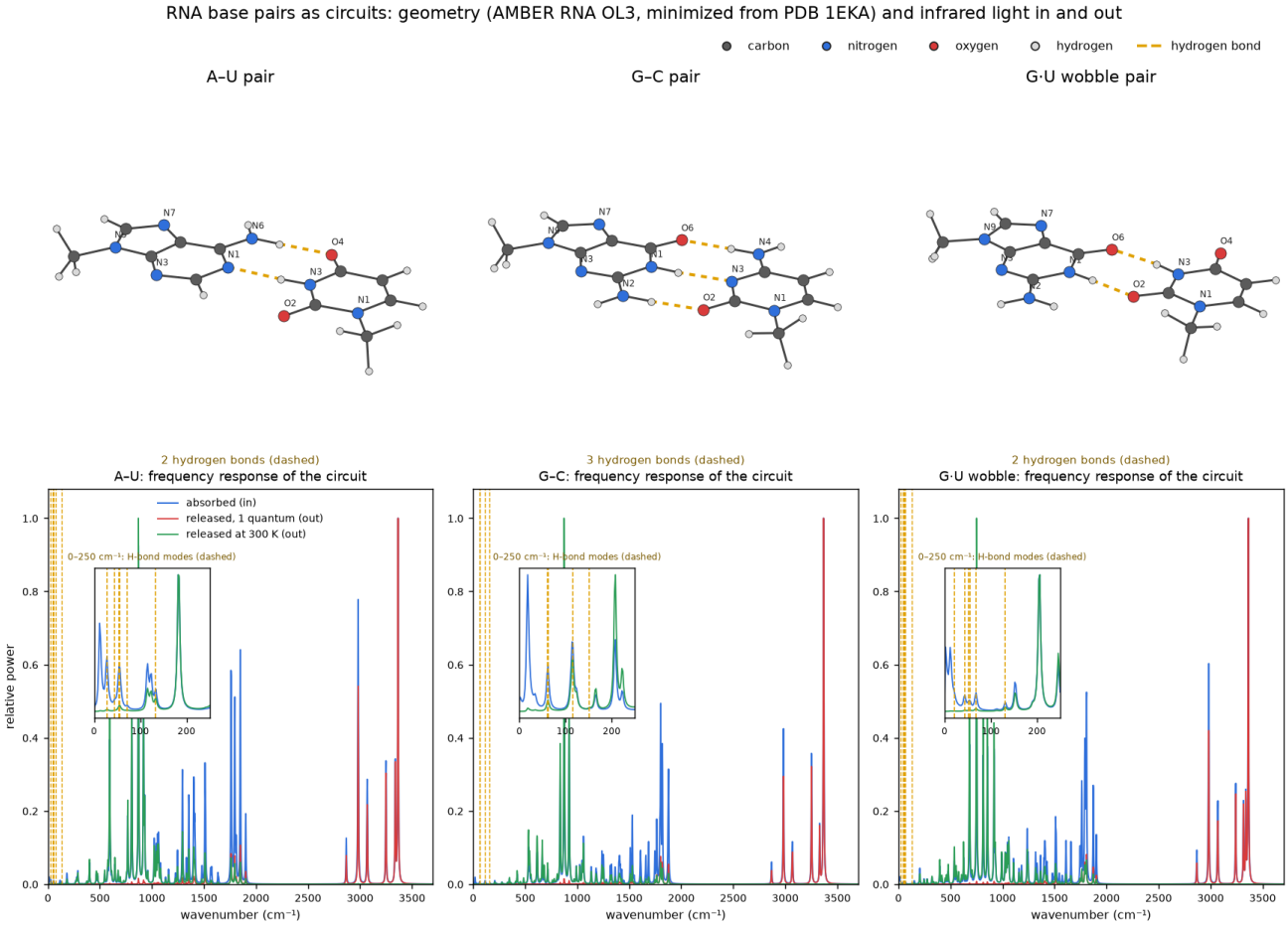
Figure 2. Model (dots) against laboratory values (triangles), with the relative difference; uracil rows in red.

Pairing and the wobble pair

pair	hydrogen bond	H···A [Å]	D···A [Å]	angle
A·U	A:N6–H61···U:O4	1.87	2.88	179°
A·U	U:N3–H3···A:N1	1.91	2.92	175°
G·C	G:N1–H1···C:N3	1.92	2.92	174°
G·C	G:N2–H21···C:O2	1.81	2.83	177°
G·C	C:N4–H41···G:O6	1.83	2.85	180°
G·U	G:N1–H1···U:O2	1.80	2.80	178°
G·U	U:N3–H3···G:O6	1.88	2.89	170°

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Modes with at least 25 % of their energy in the inter-base forces: A·U 28, 45, 53, 56, 72, 134 cm^{-1}; G·C 61, 62, 115, 152 cm^{-1}; G·U 21, 43, 52, 55, 68, 131 cm^{-1}. The wobble pair is held by the two wobble bonds only (G N1–H1···O2 U and U N3–H3···O6 G, a test); its strongest H-bond mode at 131 cm^{-1} (79 % in the H-bonds) matches A·U's 134 cm^{-1} (84 %) rather than G·C's 152 cm^{-1} (80 %). On pairing, G N1–H1 moves +30.4 cm^{-1} in G·U (+21.8 in G·C) and U N3–H3 +3.1 cm^{-1} in G·U (+26.3 in A·U). Measured H-bonded N–H stretches move down; the fixed-charge model cannot give that red shift.



A·U and G·C are Watson–Crick pairs; G·U is the wobble pair (G N1–H···O2 U and U N3–H···O6 G). Methyl groups stand in for the riboses. Harmonic model, gas phase; lines drawn 6 cm^{-1} wide.

Figure 3. Top: minimized geometry of A·U, G·C and the G·U wobble pair, hydrogen bonds dashed. Bottom: frequency response of each pair's circuit — absorbed power, release for one quantum per mode and thermal release at 300 K; insets show 0–250 cm^{-1} with the H-bond modes dashed. Lines drawn 6 cm^{-1} wide.

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Stacks (stacks/summary.csv)

stack	pairs	atoms	vibrations	lowest [cm^{-1}]	modes below 200 cm^{-1}	of which stacking	twist start → minimized
AU	1	33	93	12.5	11	0	—
AU2	2	66	192	8.5	28	4	34.50° → 33.11°
AU3	3	99	291	8.6	43	9	33.41°, 32.84° → -16.51°, 46.02°
AU4	4	132	390	6.4	59	12	30.95°, 33.61°, 30.17° → -17.49°, 43.69°, -17.49°
GC	1	35	99	17.9	11	0	—
GC2	2	70	204	16.7	28	6	35.00° → 77.18°
GC3	3	105	309	8.2	45	7	34.26°, 33.13° → 67.22°, 16.19°
GC4	4	140	414	10.0	60	13	31.23°, 34.15°, 30.39° → 56.02°, -3.13°, 54.97°
polyA2	2	66	192	8.5	28	4	34.50° → 33.11°
polyA3	3	99	291	5.9	44	9	33.44°, 32.92° → -17.03°, 23.22°
polyA4	4	132	390	4.8	60	14	30.93°, 33.65°, 30.21° → 35.96°, -32.68°, 25.35°
ApU	2	66	192	13.0	28	5	34.33° → 48.71°
GU	1	34	96	12.4	12	0	—
GU2	2	68	198	10.6	28	3	50.72° → 93.09°

Stacks: A·U (strand I AA, AAU, AAUU), G·C (CG, CGC, CGCG), poly(A)·poly(U) (AA, AAA, AAAA), ApU and the tandem wobble pairs GU2. Built stacks repeat 1EKA's GA/UC step, so AU2 and polyA2 are the same molecule. Stacking turns each single-pair line into a band and adds rigid-pair modes below 200 cm^{-1}; the lowest mode falls from 12.5 cm^{-1} (one A·U) to 6.4 (AAUU) and 4.8 cm^{-1} (AAAA), against 12.7 → 3.4 cm^{-1} for A·T in dna_vibration. With the backbone absent the free stacks leave the A-form step on minimization (AU2 rise 4.10 → 5.83 Å, GC2 twist 35.00° → 77.18°; simplified centroid and C1′–C1′ measures).

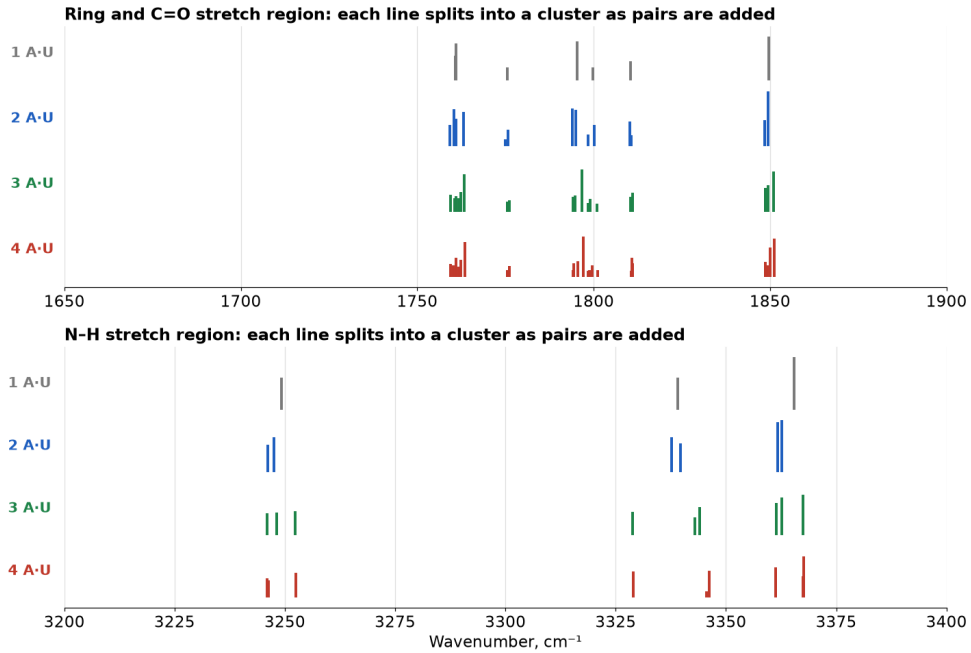


Figure 4. Ring/C=O stretch region (top) and N–H stretch region (bottom) for 1–4 stacked A·U pairs: each single-pair line splits into a cluster as pairs are added.

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Figure 5. Low-frequency vibrations of stacked A·U (top) and G·C (bottom) pairs. Thick lines: the pairs move as solid blocks against each other; thin lines: motion inside the pairs. No measured A-form RNA band was found to mark.

Validation

- Every molecule is at a true minimum and lumped by the λ/10 rule.
- The circuit's AC solution and a direct solve of the same admittance give the same absorbed power on and off resonance; the Einstein and Larmor release routes agree for every mode (rna_test.go).
- rna_test.go and ff_test.go also check the stiffness split, passivity, hydrogen bonds (the wobble pair's two by name), the modal twin, the stack parts and the analytic gradient against finite differences.
- For the 2-pair stacks the modal spectrum agrees with the direct solve and with kronspice's AC analysis of the circuit (stacks/run_output.txt).

Limits of the model

- Harmonic force field: anharmonic shifts are absent, which is why N–H stretches come out 3–6 % low.
- Fixed charges: the model cannot give the measured red shifts of H-bonded stretches on pairing.
- Gas phase, methyl caps instead of riboses, no backbone, no 2′-OH and no water: free stacks change their rise and twist on minimization.
- Stacks repeat one real step of 1EKA (GA/UC) for every sequence, a stated approximation; only GU2 has its own measured geometry.
- Electronic (UV) absorption and fluorescence are outside the vibration circuit.

Reproducing the results

```
python examples/rna_vibration/prep/prep.py RNA.OL3.xml 1eka.pdb examples/rna_vibration/molecules.json
go run ./examples/rna_vibration -out examples/rna_vibration -stacks=false
go run ./examples/rna_vibration -out examples/rna_vibration -only-stacks
go test ./examples/rna_vibration
python examples/rna_vibration/viz/make_report.py examples/rna_vibration/rna_vibration_report.pdf
```

Sources: AMBER RNA OL3 (Cornell et al., JACS 117, 5179, 1995; Pérez et al., Biophys. J. 92, 3817, 2007; Zgarbová et al., JCTC 7, 2886, 2011); Chen et al., Biochemistry 39, 8970 (2000), PDB 1EKA; Ivanov et al., Biophysical Bulletin 2(11), 14 (2002); Colarusso et al., Chem. Phys. Lett. 269, 39 (1997); Plützer et al., PCCP 3, 5466 (2001); Nowak et al., PCCP 21, 22857 (2019); Min et al., Bull. Korean

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Chem. Soc. 30, 3039 (2009); Nir et al., PCCP 4, 732 (2002); Cavaluzzi & Borer, Nucleic Acids Res. 32, e13 (2004); Varani & McClain, EMBO Rep. 1, 18 (2000); Olson et al., J. Mol. Biol. 313, 229 (2001); Wilson, Decius & Cross, Molecular Vibrations (1955).

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